

Worksheet 1 — Organising Multivariate Data

STAT 481 — Multivariate Analysis • Lecture 1 • Tue, Aug 25, 2026

1. Objective of this lecture

By the end of the hour you should be able to:

1. write a data set as an $n \times p$ array and say what one row and one column are;
2. ask any data set, in R, the three questions you ask before analysing it;
3. decide whether “this unit is extreme on one of the variables” is a useful definition of *unusual* — and prove your answer with a number.

2. The data, and how to get it in R

Package: HSAUR2 (on CRAN). **Data set:** USairpollution. **Size:** 41 US cities $\times$ 7 measurements.

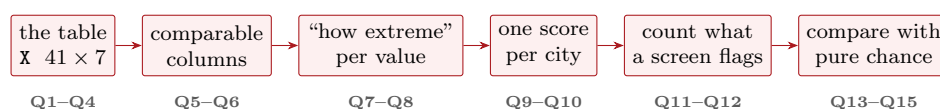
```
install.packages("HSAUR2") # ONCE, ever. Needs an internet connection.
library(HSAUR2)           # once per R session
data("USairpollution")   # copy the data set into your session
X <- USairpollution       # short name, used all hour
n <- nrow(X); p <- ncol(X)
```

S02	sulphur dioxide, $\mu\text{g}/\text{m}^3$	wind	wind speed, mph
temp	mean annual temperature, $^{\circ}\text{F}$	precip	rainfall, inches
manu	manufacturing firms, 20+ workers	predays	rainy days per year
popul	population 1970, thousands		

3. What we want to know today

The question: *which of these 41 cities is unusual — and unusual how?*

Not today. Today we settle a smaller one: **is the instinctive method any good?** — the instinctive method being *take the variables one at a time and pick out the extreme values*. The questions below are the workflow that settles it; each produces the material the next one needs.



4. The questions

I Stage 1 — the table

Q1. How many units and how many variables does X have?

R code

Q2. Which single function tells you the type of every column at once?

R code

Q3. Print the first rows. The leftmost column holds city names — is it an eighth variable?

guess

Q4. Get one number: Phoenix's rainfall.

R code

| Stage 2 — making the seven columns comparable

Q5. The columns are in degrees Fahrenheit, in thousands of people, in inches. Name one way to put them on a common footing — and say what it assumes.

guess

Q6. Replace every value by its rank *inside its own column*.

R code

| Stage 3 — “how extreme” — one number per value

Rank, precisely. Sort *one column* on its own, smallest to largest. The **rank** r of a value is its position in that order: $r = 1$ is the smallest of the 41 values *in that column*, $r = n = 41$ the largest. Tied values share the average of the positions they fill, so in general $r_i = \#\{j : x_j < x_i\} + \frac{1}{2}(1 + \#\{j : x_j = x_i\})$ — count how many are strictly below you, then add half of one-plus-your-own-tie-group; for $x = (7, 2, 9, 2)$ this gives $(3, 1.5, 4, 1.5)$. *Why ranks:* they carry no units, so “12th of 41” on `temp` and “17th of 41” on `popul` can be compared where 55 degrees and 608 thousand people cannot. Ranks are formed within a column, never across columns — this is what `rank()` returned in Q6. *Careful:* your printed notes rank the other way round ($1 = \text{largest}$), so always say which end you mean.

Q7. $r = 1$ and $r = 41$ are both extreme; $r = 21$ is the most ordinary a value can be. Write a formula $e(r)$, using only r and n , that measures how extreme a value is and returns the *same* number at both ends of the column.

guess

Q8. Apply your formula to the whole 41×7 table at once.

R code

| Stage 4 — one score per city

Q9. Reduce each city's seven scores to one: its best claim to being extreme.

R code

Q10. Does a *large* value of that score mean the city is more unusual, or less?

guess

| Stage 5 — measuring the instinctive method

Q11. A city is FLAGGED if any of its 7 values is among the **3 most extreme** in its column (either end). Out of 41, how many? Commit to a number. guess

My prediction: of 41

Q12. Count them.

R code

| Stage 6 — is that a lot? — comparing with pure chance

Q13. If the seven columns carried *unrelated* information, what proportion of the 41 would be flagged? Formula in n, p, k , then the number. guess

Q14. Observed _____% against chance _____%. Which is bigger — and what does that say about the seven *columns*? guess

Q15. Chicago has the most S02 and the largest popul. On a plot of temp against precip, where does it land? guess

5. Conclusion — write it in your own words

A screen that flags a city whenever any one of its 7 measurements is among the 3 most extreme in its column flags _____ of 41 cities = _____%, against _____% for unrelated columns.

Therefore “this city is extreme on one of the variables” is

And the columns are

The opening question is still open, because every step above treated the columns

6. Next lecture

Thursday, Aug 27 — Summary Statistics. Something holds the seven columns together and we have no name for it. Thursday it gets a name, a symbol and a 7×7 table, and the 287 numbers of **X** collapse into a handful. *Bring this sheet:* the two percentages you wrote in the conclusion are what Thursday explains.